

Homework Assignment 3: Criticality in the S_N Approximation

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August 15, 2026

The method

All three questions solve the same monoenergetic, steady-state problem,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{1}{2} \left[\Sigma_s \phi + \frac{1}{k} \nu \Sigma_f \phi \right], \quad \phi(x) = \int_{-1}^1 \psi(x, \mu) d\mu \quad (1)$$

with the same three ingredients: a Gauss–Legendre quadrature in angle, diamond differencing in space with a negative-flux fixup, and the Bell & Glasstone k iteration. Lengths are mean free paths in Questions 3 and 4 and centimetres in Question 5.

Angle. The N ordinates $\{\mu_m\}$ and weights $\{w_m\}$ are the Gauss–Legendre nodes on $[-1, 1]$, so that $\mu_{N-m+1} = -\mu_m$, $\sum_m w_m = 2$, and

$$\phi_i = \sum_{m=1}^N w_m \psi_{i,m} \quad (2)$$

Space, slab. Integrating over cell i and closing with the diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i+1/2,m} + \psi_{i-1/2,m})$ gives the sweep

$$\psi_{i,m} = \frac{2|\mu_m| \psi_{\text{in},m} + \frac{1}{2} S_i \Delta x}{\Sigma_t \Delta x + 2|\mu_m|}, \quad \psi_{\text{out},m} = 2\psi_{i,m} - \psi_{\text{in},m} \quad (3)$$

swept from $x = a/2$ inwards for $\mu_m < 0$ and from $x = 0$ outwards for $\mu_m > 0$.

Space, sphere. The spherical equation carries an angular redistribution term,

$$\mu \frac{\partial \psi}{\partial r} + \frac{1 - \mu^2}{r} \frac{\partial \psi}{\partial \mu} + \Sigma_t \psi = \frac{S}{2} \quad (4)$$

whose conservative form integrates over the cell to

$$\mu_m \left(A_{i+\frac{1}{2}} \psi_{i+\frac{1}{2},m} - A_{i-\frac{1}{2}} \psi_{i-\frac{1}{2},m} \right) + \frac{A_{i+\frac{1}{2}} - A_{i-\frac{1}{2}}}{w_m} \left(\alpha_{m+\frac{1}{2}} \psi_{i,m+\frac{1}{2}} - \alpha_{m-\frac{1}{2}} \psi_{i,m-\frac{1}{2}} \right) + \Sigma_t V_i \psi_{i,m} = \frac{S_i}{2} V_i \quad (5)$$

with $A = 4\pi r^2$ and $V_i = \frac{4\pi}{3}(r_{i+1/2}^3 - r_{i-1/2}^3)$. This is (3) with a second outgoing face, in angle, closed by the same diamond relation $\psi_{i,m} = \frac{1}{2}(\psi_{i,m+1/2} + \psi_{i,m-1/2})$. The angular coefficients follow from demanding that a flat flux solve (5) exactly:

$$\alpha_{\frac{1}{2}} = 0, \quad \alpha_{m+\frac{1}{2}} = \alpha_{m-\frac{1}{2}} - w_m \mu_m, \quad \alpha_{N+\frac{1}{2}} = 0 \quad \text{since} \quad \sum_m w_m \mu_m = 0 \quad (6)$$

The recursion needs a starting value $\psi_{i,1/2}$ at $\mu_{1/2} = -1$, where $1 - \mu^2 = 0$ and the angular derivative drops out; what remains, $-\partial\psi/\partial r + \Sigma_t\psi = S/2$, is the plain slab sweep at $|\mu| = 1$, run inwards from the vacuum boundary.

One cell solve for both geometries. Equations (3) and (5) are the same equation over a different number of faces, and it is worth writing that form down once because it is what the code implements. Let a cell carry a list of outgoing faces f , each contributing $a_{\text{out}}^f \psi_{\text{out}}^f - a_{\text{in}}^f \psi_{\text{in}}^f$ to the balance:

$$\sum_f \left(a_{\text{out}}^f \psi_{\text{out}}^f - a_{\text{in}}^f \psi_{\text{in}}^f \right) + \Sigma_t V \psi = \frac{1}{2} S V \quad (7)$$

Closing *every* face with the same diamond relation $\psi_{\text{out}}^f = 2\psi - \psi_{\text{in}}^f$ and collecting ψ gives one formula for any geometry:

$$\psi = \frac{\frac{1}{2} S V + \sum_f \left(a_{\text{out}}^f + a_{\text{in}}^f \right) \psi_{\text{in}}^f}{\Sigma_t V + 2 \sum_f a_{\text{out}}^f} \quad (8)$$

The two geometries differ only in what they put in the list:

	faces	a_{out}	a_{in}
slab	space	$ \mu_m $	$ \mu_m $
sphere	space	$ \mu_m A_{\text{out}}$	$ \mu_m A_{\text{in}}$
	angle	$\frac{A_{i+1/2} - A_{i-1/2}}{w_m} \alpha_{m+\frac{1}{2}}$	$\frac{A_{i+1/2} - A_{i-1/2}}{w_m} \alpha_{m-\frac{1}{2}}$

with $V = \Delta x$ in the slab. **A face's two coefficients differ only where it is weighted differently on the way in and on the way out** — the two areas of a spherical shell, the two α 's of an angular bin — which is why the slab's are equal and the distinction looks empty there. Substituting the slab row into (8) returns (3) exactly, and the sphere's two rows return (5). The negative-flux fixup below is the statement that a face whose ψ_{out} has been clamped to zero drops its a_{out} from both sums in (8), keeping only its inflow.

Fixup. Whenever a diamond outgoing flux $2\psi_{i,m} - \psi_{\text{in},m}$ comes out negative it is set to zero and the cell balance is re-solved with it held there, in angle as well as in space. This never occurs in Questions 3–5: the fission source is proportional to the flux itself, so no cell is ever starved, and a critical system is thin enough that $\Sigma_t \Delta x / |\mu_m| < 2$ at every ordinate.

Boundary conditions. Reflective at the symmetry point, $\psi_{1/2,m} = \psi_{1/2,m'}$ with $\mu_{m'} = -\mu_m$; vacuum at the outer face, $\psi_{I+1/2,m} = 0$ for $\mu_m < 0$.

Eigenvalue. Outer iteration l fixes the fission source $Q_{f,i}^{(l)} = \nu \Sigma_{f,i} \phi_i^{(l)} / k^{(l)}$; inner iteration n sweeps $S_i = \Sigma_{s,i} \phi_i^{(n)} + Q_{f,i}^{(l)}$ until ϕ settles; then

$$k^{(l+1)} = k^{(l)} \frac{P^{(l+1)}}{P^{(l)}}, \quad P^{(l)} = \sum_i \nu \Sigma_{f,i} \phi_i^{(l)} V_i \quad (9)$$

Dividing the fission source by k makes a solution exist at *every* system size, so criticality becomes a root search: **brentq** on $k(\text{size}) - 1$, which costs 8 evaluations of k .

The meaning of c . Questions 3 and 4 specify only $c = (\Sigma_s + \nu\Sigma_f)/\Sigma_t$. At $k = 1$ the source bracket collapses to $c\Sigma_t\phi$ whatever the split between scattering and fission, so the critical size depends on c alone and the code takes the simplest split, $\Sigma_t = 1$, $\Sigma_s = 0$, $\nu\Sigma_f = c$. Question 5 has no such freedom and uses the tabulated cross sections directly.

1 Critical radius of a reflected sphere

Question 1 is the one *diffusion* question of this assignment — the S_N machinery above belongs to Questions 3–5 — and the only one with two materials. A multiplying core of radius R_C is wrapped in $d \in \{1, 2, 3, 10\}$ mean free paths of reflector, and the cross sections are again the Question 5 table of Assignment 1.

The two regions

In both regions the substitution $u(r) = r\phi(r)$, which uses $\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \frac{1}{r} \frac{d^2 u}{dr^2}$, turns the spherical diffusion equation into a plane one:

$$\frac{d^2 u}{dr^2} - \frac{(1-c)\Sigma_t^2}{D_0} u = 0 \quad (10)$$

The core has $c_C > 1$, so the coefficient is negative and the solution oscillates; finiteness at the origin selects the sine. The reflector has $c_R \leq 1$ and decays. With

$$B = k_0(c_C)\Sigma_{t,C}, \quad \kappa = \frac{\Sigma_{t,R}}{\nu_0(c_R)}, \quad R_{\text{ext}} = R_R + \frac{z_0(c_R)}{\Sigma_{t,R}} \quad (11)$$

the two profiles are

$$\phi_C(r) = A \frac{\sin(Br)}{r}, \quad \phi_R(r) = F \frac{\sinh[\kappa(R_{\text{ext}} - r)]}{r} \quad (12)$$

k_0 and ν_0 are Case's discrete eigenvalue on its two branches, the same roots Assignment 1 used: $c \arctan(k_0) = k_0$ above $c = 1$ and $\text{artanh}(1/\nu_0)/(1/\nu_0) = 1/c$ below it.

Matching, and the one criticality equation

The interface at $r = R_C$ carries a flux relation $\phi_R = g\phi_C$ and continuity of the net current $J = -D d\phi/dr$. Since $\phi = u/r$ gives $\phi'/\phi = u'/u - 1/r$, and the two logarithmic derivatives are $u'_C/u_C = B \cot(BR_C)$ and $u'_R/u_R = -\kappa \coth(\kappa L)$ with $L = R_{\text{ext}} - R_C$, all three approximations reduce to a single transcendental equation for R_C :

$$\boxed{D_C B \cot(BR_C) + g D_R \kappa \coth(\kappa L) - \frac{D_C - g D_R}{R_C} = 0} \quad (13)$$

with $\kappa L = [d + z_0(c_R)]/\nu_0(c_R)$, so the reflector enters only through its *optical* thickness. The last term has no slab counterpart: ϕ falls as $1/r$ even where u is flat, so a jump in D at a curved interface drives a current by itself. Equation (13) runs from $+\infty$ at $R_C \rightarrow 0$ to $-\infty$ at $R_C = \pi/B$, the unreflected limit, so the fundamental root is always bracketed by $(0, \pi/B)$ and **brentq** finds it in one call.

The three approximations differ only in what they put into D_0 , z_0 and g :

	(a) continuous classic	(b) continuous asymptotic	(c) discontinuous asymptotic
$D_0(c)$	$1/3$	$ 1 - c \nu_0^2$	$ 1 - c \nu_0^2$
relaxation rate	$\sqrt{3 1 - c }$	$k_0(c)$ or $1/\nu_0(c)$	$k_0(c)$ or $1/\nu_0(c)$
$z_0(c)$	$2/3$	$0.710446 [1 + q(1 - c)^2]/c$	$0.710446 [1 + q(1 - c)^2]/c$
$g = \phi_R/\phi_C$	1	1	$\mu_0(c_C)/\mu_0(c_R)$

Table 1: Only the last row makes (c) different from (b). Since $\mu_0 = 1/2$ for every medium in the classic theory, (a) is automatically continuous, and with no reflector at all (b) and (c) are the same bare sphere.

Zimmerman’s μ_0 . The discontinuity comes from Zimmerman (UCRL-82792), who writes the partial currents of the asymptotic mode as $\phi_{\pm} = \frac{1}{2}\mu_0\phi_0 \pm \frac{1}{2}\phi_1$, so that $\mu_0\phi_0 = J_+ + J_-$ is the total traffic across a surface and

$$\mu_0(c) = \frac{c\nu_0^2}{2} \ln\left(\frac{\nu_0^2}{\nu_0^2 - 1}\right) \xrightarrow{c>1} \frac{c}{2k_0^2} \ln(1 + k_0^2) \quad (14)$$

the second form being the exact analytic continuation through $\nu_0 = i/k_0$. It is not an ansatz: integrating the discrete mode $\psi(\mu) \propto (\nu_0 - \mu)^{-1}$ over the two half-ranges reproduces it identically, and it returns the two textbook limits $\mu_0 = 1/2$ for an isotropic flux and $\mu_0 = 1$ for a one-way beam. Matching the traffic rather than the flux, $\mu_0(c_C)\phi_C = \mu_0(c_R)\phi_R$, is what makes ϕ itself jump. Nothing is lost at the interface — the same current crosses both faces.

material	c	mfp [cm]	k_0 or $1/\nu_0$	D_0	D [cm]	μ_0	z_0
Pu-239	1.5000	3.064	1.45110	0.23745	0.7275	0.40363	0.47127
U-235	1.3000	3.064	1.05708	0.26847	0.8225	0.43639	0.54552
H ₂ O	0.9000	3.064	0.52543	0.36222	1.1097	0.52660	0.78923
Fe	0.9980	4.300	0.07740	0.33387	1.4356	0.50050	0.71187
Na	1.0000	11.578	0.00000	0.33333	3.8595	0.50000	0.71045

Table 2: Asymptotic parameters of the five materials. Sodium is a pure scatterer, $\Sigma_c = \Sigma_f = 0$, so $c_R = 1$ exactly: ν_0 diverges, $\kappa = 0$, and the reflector solution degenerates to u_R linear in r with $\kappa \coth(\kappa L) \rightarrow 1/L$. The code takes that limit explicitly rather than dividing out a near-zero κ .

Results

The bare spheres of Assignment 1 Question 5, which every reflected radius below is measured against:

core	classic R_c [cm]	asympt. R_c [cm]	classic M_c [kg]	asympt. M_c [kg]
Pu-239	5.8163	5.1890	12.940	9.188
U-235	8.1031	7.4339	42.345	32.696

Table 3: Bare reference. Theory (a) is compared with the classical column, (b) and (c) with the asymptotic one.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	5.3159	4.9484	4.5400	9.879	7.968	6.154
Water	2	5.0855	4.7987	4.3823	8.649	7.267	5.535
Water	3	5.0153	4.7499	4.3321	8.296	7.048	5.347
Water	10	4.9814	4.7244	4.3061	8.129	6.935	5.251
Iron	1	5.4212	4.9575	4.6450	10.478	8.013	6.591
Iron	2	5.1056	4.7364	4.4182	8.753	6.988	5.672
Iron	3	4.9506	4.6237	4.3059	7.979	6.501	5.250
Iron	10	4.6790	4.4188	4.1070	6.737	5.674	4.556
Sodium	1	6.4397	5.7159	5.5171	17.562	12.281	11.044
Sodium	2	6.3134	5.6405	5.4318	16.549	11.801	10.540
Sodium	3	6.2505	5.6022	5.3892	16.059	11.563	10.293
Sodium	10	6.1315	5.5286	5.3080	15.159	11.113	9.835

Table 4: Pu-239 core: critical radius in cm and critical mass in kg, under (a) continuous classic, (b) continuous asymptotic and (c) discontinuous asymptotic diffusion. Bare: 5.8163 cm / 12.940 kg classic, 5.1890 cm / 9.188 kg asymptotic.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	7.2501	6.9113	6.5228	30.330	26.273	22.087
Water	2	6.9185	6.6731	6.2694	26.355	23.649	19.611
Water	3	6.8146	6.5937	6.1865	25.186	22.816	18.844
Water	10	6.7638	6.5520	6.1432	24.627	22.385	18.451
Iron	1	7.2811	6.8392	6.5656	30.720	25.460	22.525
Iron	2	6.8002	6.4625	6.1789	25.027	21.480	18.775
Iron	3	6.5545	6.2644	5.9804	22.411	19.565	17.023
Iron	10	6.1125	5.8970	5.6199	18.176	16.320	14.126
Sodium	1	8.4264	7.7951	7.6194	47.618	37.697	35.205
Sodium	2	8.2322	7.6564	7.4692	44.401	35.721	33.164
Sodium	3	8.1328	7.5844	7.3919	42.811	34.722	32.145
Sodium	10	7.9401	7.4425	7.2414	39.840	32.809	30.221

Table 5: U-235 core, same columns. Bare: 8.1031 cm / 42.345 kg classic, 7.4339 cm / 32.696 kg asymptotic.

Question 1: critical core radius of the reflected spheres

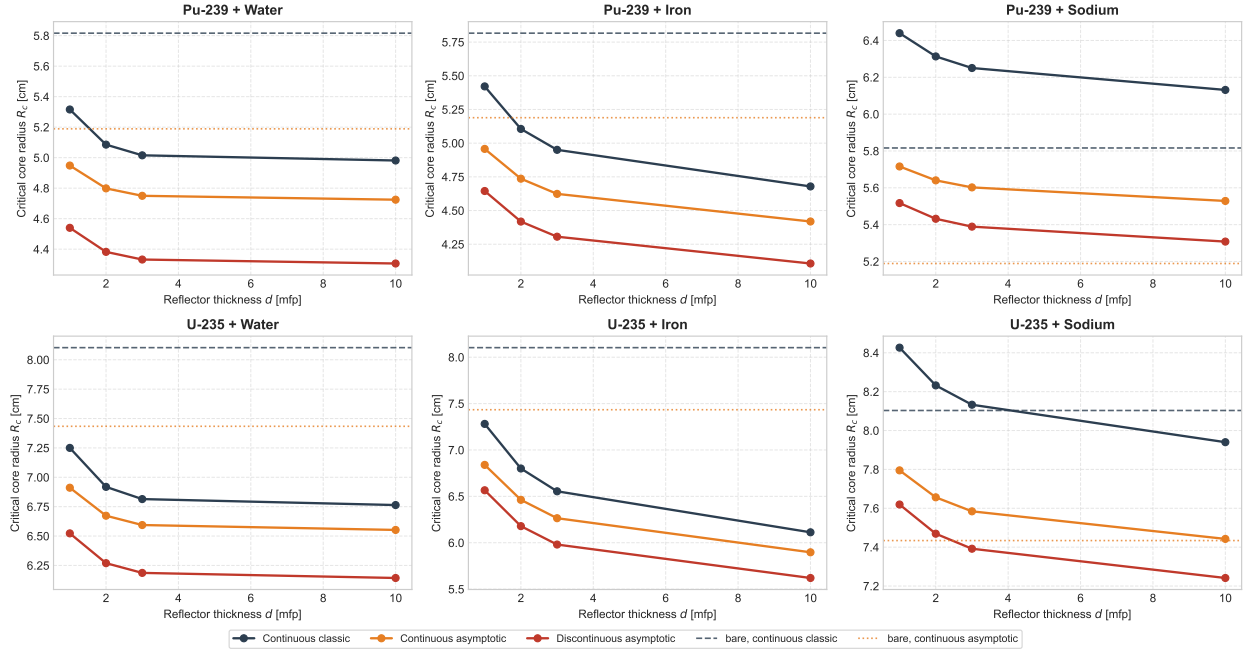


Figure 1: Critical core radius against reflector optical thickness. Dashed and dotted lines are the two bare references. Water saturates by $d \approx 3$ mfp; iron is still improving at $d = 10$; sodium sits above its bare line throughout.

Question 1: critical flux profiles behind 3 mfp of reflector

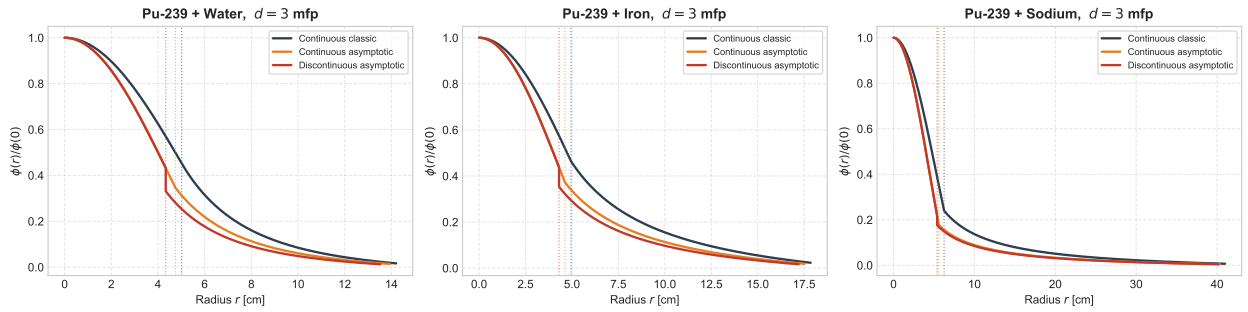


Figure 2: Critical flux across core and reflector for the Pu-239 core behind 3 mfp, normalised to $\phi(0) = 1$; dotted verticals mark each theory's interface. The discontinuous curve drops by the factor g at the interface and is continuous in current, not in flux. Note the horizontal scale of the sodium panel: 3 mfp of sodium is 34.7 cm.

What the three approximations say

The reflector saving. Water and iron behave as reflectors should. Both cores lose 12–17% of their radius behind water and 12–24% behind iron by theory (c), which because the mass goes as R^3 is a factor of nearly two in critical mass: Pu-239 falls from 9.188 kg bare to 4.556 kg behind 10 mfp of iron. The two reflectors saturate differently, and the reason is c_R . Water absorbs ($c_R = 0.90$, $\nu_0 = 1.90$), so a neutron more than about two mean free paths deep never returns and the water columns are flat past $d = 3$. Iron barely absorbs ($c_R = 0.998$, $\nu_0 = 12.9$), so its return keeps improving out to $d = 10$ and beyond — iron starts as the worse reflector at $d = 1$ and ends the better one at $d = 10$.

The ordering of the three theories. At every entry $R_c^{(a)} > R_c^{(b)} > R_c^{(c)}$. The first gap is inherited from the bare problem: the asymptotic $D_0(c_C)$ and $k_0(c_C)$ already give a smaller sphere than $D_0 = 1/3$ does, by 11% for Pu-239. The second is the jump itself. Every pair here has $\mu_0(c_C) < \mu_0(c_R)$, so $g < 1$ — 0.766 for Pu-239 behind water, 0.873 for U-235 behind iron — because the multiplying core has the shorter relaxation length and hence the more forward-peaked angular flux, and so carries more current per unit density. A g below one multiplies down both the leakage term and the curvature term of (13), which is why (c) always predicts the smallest core, by a further 8–9% for water and iron.

The sodium columns are diffusion theory failing. Every sodium radius above is *larger* than the corresponding bare sphere, which no reflector can do. The condition is being solved correctly — (13) was checked against an independent two-region spherical finite-volume k -eigenvalue solve, 3000 cells with harmonic-mean face diffusion coefficients across the material jump, and the two agree to better than 0.4% on exactly these cases. The fault is in the model. Splitting the interface leakage of the Pu-239 core at $d = 10$ mfp into its two parts, $J/\phi = D_R/R_c + D_R/L$:

reflector	D_R [cm]	shell [cm]	geometric D_R/R_c	far-face D_R/L	total vs. bare 0.5592
Water	1.021	30.6	0.2050	0.1826	0.3876
Iron	1.433	43.0	0.3063	0.0381	0.3444
Sodium	3.859	115.8	0.6295	0.0312	0.6607

Table 6: Leakage ratio at the interface, in cm^{-1} , classic theory. Sodium’s far-face return is the best of the three — 10 mfp of a pure scatterer really does send almost everything back — but its geometric term alone exceeds what a bare vacuum surface leaks.

The geometric term is the current driven by the $1/r$ spreading of $\phi = u/r$, and it is harmless only while $D_R \ll R_c$. Sodium’s $\Sigma_t = 0.086368 \text{ cm}^{-1}$ is a mean free path of 11.58 cm against a core radius near 6 cm: a neutron entering the reflector travels two core radii before its first collision, which is precisely the regime Fick’s law expands away from. None of the three approximations repairs it, because a pure scatterer has $D_0 = 1/3$ in all of them and $D_R = 3.86 \text{ cm}$ throughout. Measured against its own bare sphere at $d = 10$, (a) is +5.4%, (b) +6.6% and (c) +2.3% — only the Zimmerman jump helps, and only because $g = 0.807$ multiplies the offending term. Transport theory would not make this mistake: the shell is optically thick and purely scattering, so its true albedo is high and the true critical radius must lie below the bare 5.1890 cm.

3 Critical slab half-thickness by S_N

Reflective at $x = 0$, vacuum at $x = a/2$, 200 cells. The reference column is Assignment 1's exact-transport relation $a/2 = \frac{\pi}{2} \nu_0 - z_0$, evaluated from Case's tabulated ν_0 and z_0 .

c	order	$a/2$ [mfp]	exact transport [mfp]	departure
1.2	S_2	1.48499	1.28981	+15.13%
1.2	S_4	1.31855	1.28981	+2.23%
1.2	S_6	1.29861	1.28981	+0.68%
1.2	S_{10}	1.29218	1.28981	+0.18%
1.5	S_2	0.78001	0.60762	+28.37%
1.5	S_4	0.64949	0.60762	+6.89%
1.5	S_6	0.62089	0.60762	+2.18%
1.5	S_{10}	0.60904	0.60762	+0.23%
1.8	S_2	0.54291	0.39314	+38.09%
1.8	S_4	0.43875	0.39314	+11.60%
1.8	S_6	0.41014	0.39314	+4.32%
1.8	S_{10}	0.39459	0.39314	+0.37%

Table 7: Critical half-thickness in mean free paths. The departure is entirely angular: the same radii are reproduced to seven digits on meshes from 50 to 800 cells.

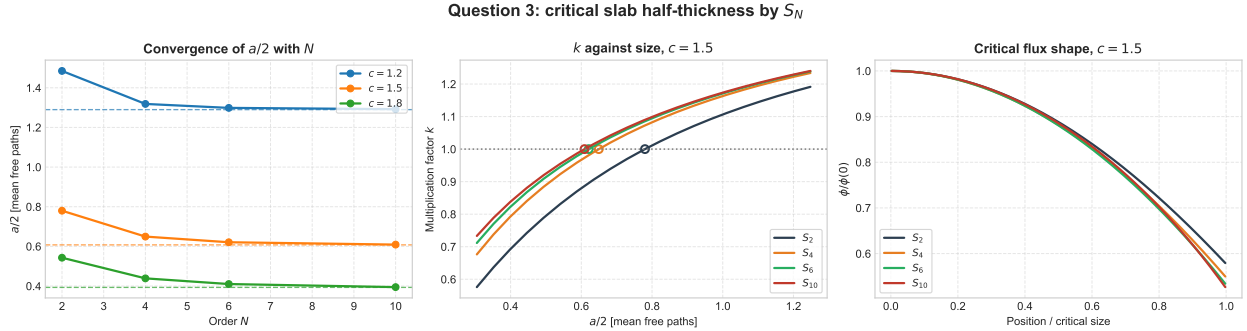


Figure 3: Left: convergence of $a/2$ with N , dashed lines the exact-transport values. Centre: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings. Right: the normalised critical flux, which barely moves with N — the order affects the boundary, not the interior.

The exact one-speed value at $c = 1.5$ is 0.605055 mfp, the Sood *et al.* Pu-239 slab (1.853722 cm at $\Sigma_t = 0.32640 \text{ cm}^{-1}$, the same table Assignment 1 Question 5 uses). The S_N sequence walks down onto it monotonically: 0.609042 at S_{10} , 0.606407 at S_{16} , 0.605631 at S_{24} , 0.605195 at S_{48} .

P_N against S_{N+1}

In slab geometry the two are algebraically the same interior equations. Expanding ψ in $N + 1$ Legendre moments and truncating gives the same $N + 1$ coupled ODEs as collocating at the $N + 1$ Gauss–Legendre points, because a Gauss–Legendre rule of order $N + 1$ integrates polynomials up to degree $2N + 1$ exactly. **Any difference between P_N and S_{N+1} is therefore a difference of boundary condition**, and it is largest at low N where the boundary is the whole story:

- S_{N+1} sets $\psi(a/2, \mu_m) = 0$ on each incoming ordinate — the *Mark* condition.
- P_N needs conditions on moments instead. *Marshak*'s, $\int_0^1 P_{2j+1}(\mu)\psi d\mu = 0$, weight the whole incoming half-range; Mark's pin the same isolated directions S_{N+1} does.

With Mark conditions the two agree; with Marshak they do not. Concretely at $c = 1.5$: S_2 above gives $a/2 = 0.78001$, and P_1 with the Mark extrapolation $l_0 = 1/\sqrt{3}$ gives $\arctan(1/Bl_0)/B = 0.78001$ — the same five digits — while P_1 with the Marshak $l_0 = 2/3$ gives 0.72348, a 7% spread between two methods that share every interior equation.

A second source of difference is that S_N is a collocation method: it satisfies the equation exactly along N directions and says nothing between them. At the vacuum face the leaked current $\sum_{\mu_m > 0} w_m \mu_m \psi_m$ is that of a flux strongly peaked towards $\mu = 1$, which a coarse quadrature underestimates — so the slab has to be made *thicker* than it should be to stay critical, and the table comes down from above.

All of that is a statement about *slabs*. In the sphere the equivalence fails at the interior equations themselves: the angular derivative $\frac{1-\mu^2}{r} \frac{\partial \psi}{\partial \mu}$ couples the Legendre moments in a different pattern from the way the α recursion of (5) couples the ordinates, so P_1 and S_2 are genuinely different approximations there rather than the same one under two boundary conditions. At $c = 1.5$, S_2 gives $R_c = 1.607$ mean free paths where P_1 with the Mark extrapolation gives 1.919 and with Marshak's 1.822 — a spread no choice of boundary condition can close.

4 Critical sphere radius by S_N

Reflective at $r = 0$, vacuum at $r = R$, 100 cells, with the angular redistribution term of (5). The reference is Assignment 1's $\Sigma_t R_c = \pi \nu_0 - z_0$.

c	order	R_c [mfp]	exact transport [mfp]	departure
1.2	S_2	3.06060	3.17204	−3.51%
1.2	S_4	3.14631	3.17204	−0.81%
1.2	S_6	3.15991	3.17204	−0.38%
1.2	S_{10}	3.16728	3.17204	−0.15%
1.5	S_2	1.60725	1.69010	−4.90%
1.5	S_4	1.66739	1.69010	−1.34%
1.5	S_6	1.67934	1.69010	−0.64%
1.5	S_{10}	1.68595	1.69010	−0.25%
1.8	S_2	1.11753	1.18295	−5.53%
1.8	S_4	1.16413	1.18295	−1.59%
1.8	S_6	1.17412	1.18295	−0.75%
1.8	S_{10}	1.17971	1.18295	−0.27%

Table 8: Critical radius in mean free paths. Note the sign: the sphere converges from *below*, the slab from above.

Cross-check between the two geometries. Both drive the asymptotic flux to zero at the same extrapolated surface, so $a/2 + z_0 = \frac{\pi}{2} \nu_0$ and $R_c + z_0 = \pi \nu_0$, and therefore

$$z_0 = R_c - 2(a/2) \tag{15}$$

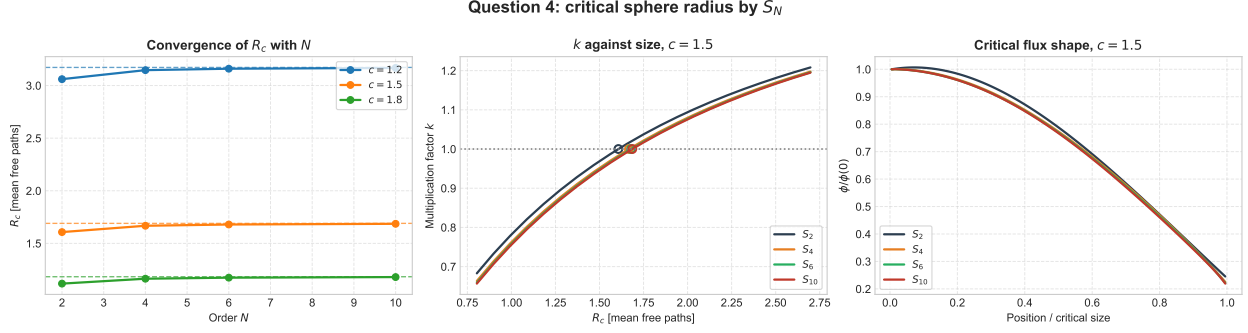


Figure 4: The same three panels for the sphere. The S_2 curve is the outlier in all three: in curvilinear geometry S_2 is not the P_1 equation, and the angular redistribution term it misrepresents is a loss from the interior directions.

which lets the two tables above check each other without any further calculation:

c	from S_2	from S_4	from S_6	from S_{10}	Case's table
1.2	0.091	0.509	0.563	0.583	0.592
1.5	0.047	0.368	0.438	0.468	0.475
1.8	0.032	0.287	0.354	0.391	0.397

Table 9: Extrapolation distance z_0 implied by the two geometries, in mean free paths. At S_{10} it is 1.5% below Case's value at all three c — the same shortfall three times, which is the relation's own asymptotic error. At S_2 it nearly vanishes, which is the sharpest statement of what low-order S_N gets wrong: the boundary, not the interior.

5 Critical mass of U-235 and Pu-239 by S_N

The cross sections are the Question 5 table of Assignment 1 — $\Sigma_t = 0.32640 \text{ cm}^{-1}$ for both, giving $c = 1.500$ for Pu-239 and $c = 1.300$ for U-235 — and the critical mass follows from the converged radius as $M_c = \frac{4}{3}\pi R_c^3 \rho$. Here $\Sigma_s > 0$, so the inner iteration does real work: about ten sweeps per outer, against one for the c -only problems above. The orders are S_2 to S_{16} rather than the S_2 – S_{10} of the previous questions, since the mass goes as the cube of the radius and is the more demanding quantity.

Diffusion overestimates both radii, and by more for Pu-239 than for U-235: the surface sits about two mean free paths from the centre in both cases, but the higher c makes the Pu-239 flux fall off faster and its surface angular flux more forward-peaked, which is exactly what a diffusion approximation cannot represent. S_{16} recovers the difference to within 0.3% of the transport-theory radius.

material	order	c	R_c [cm]	M_c [kg]	asympt. diffusion R_c [cm]	departure
Pu-239	S_2	1.500	4.9242	7.852	5.1890	-5.10%
Pu-239	S_4	1.500	5.1084	8.767	5.1890	-1.55%
Pu-239	S_8	1.500	5.1586	9.028	5.1890	-0.59%
Pu-239	S_{16}	1.500	5.1730	9.104	5.1890	-0.31%
U-235	S_2	1.300	7.1218	28.748	7.4339	-4.20%
U-235	S_4	1.300	7.3514	31.619	7.4339	-1.11%
U-235	S_8	1.300	7.4072	32.345	7.4339	-0.36%
U-235	S_{16}	1.300	7.4231	32.553	7.4339	-0.15%

Table 10: Critical radius and mass, against the asymptotic diffusion result of Assignment 1 Question 5. The mass is the cube of the radius, so the 5% radius error of S_2 is a 14% error in mass.

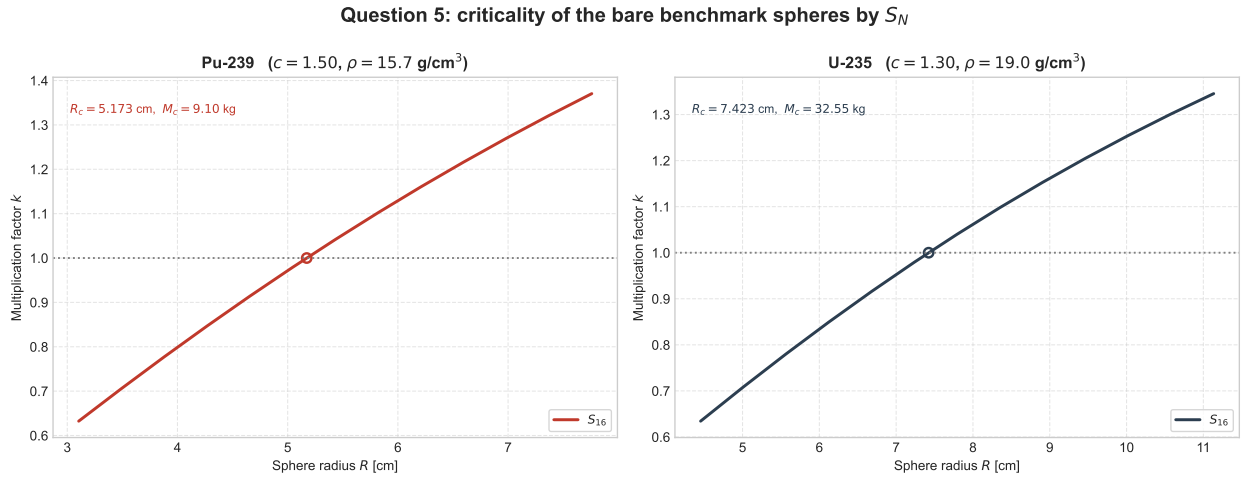


Figure 5: k against sphere radius at S_{16} , with the $k = 1$ crossing that fixes the critical mass.